

(OR)

- b. Using power method, find all the Eigen values of $A = \begin{bmatrix} 5 & 0 & 1 \\ 0 & -2 & 0 \\ 1 & 0 & 5 \end{bmatrix}$.

29. a. Estimate the value of $f(22)$ and $f(42)$ from the following data

x	20	25	30	35	40	45
f(x)	354	332	291	260	231	204

(OR)

- b.i. Find the polynomial $f(x)$ by using Lagrange's formula and hence find $f(3)$ for the following data

x	0	1	2	5
y	2	3	12	147

(8 Marks)

- ii. Find the divided difference table of the function $f(x) = x^3 + x + 2$ for the arguments 1, 3, 6, 11. (4 Marks)

30. a. Find the first and second derivative of $\sqrt{x}$ at $x = 15$ from the table below

x	15	17	19	21	23	25
$\sqrt{x}$	3.873	4.123	4.359	4.583	4.796	5.000

(OR)

- b. Evaluate $\int_0^1 \frac{1}{1+x^2}$ using Simpson's $\frac{1}{3}$ rule with $h = \frac{1}{4}$ and Simpson's $\frac{3}{8}$ rule with $h = \frac{1}{6}$.

31. a. Apply fourth order Runge Kutta method to find an approximate value of y when $x = 0.2$ given that $y' = x + y$, $y(0) = 1$, taking $h = 0.1$.

(OR)

- b.i. Using Euler's method find y for $x = 0.1$ given $\frac{dy}{dx} = \frac{y-x}{y+x}$, $y(0) = 1$, taking $h = 0.05$. (4 Marks)

- ii. Solve $y' = \frac{1}{2}(1+x^2)y^2$, $y(0) = 1$, $y(0.1) = 1.06$, $y(0.2) = 1.12$, $y(0.3) = 1.21$ and compute $y(0.4)$ using Milne's predictor corrector method. (8 Marks)

32. a. By iteration method, solve $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ over a square region of side 4, satisfying the boundary conditions

- (i) $u(0, y) = 0$
- (ii) $u(4, y) = 12 + y$
- (iii) $u(x, 0) = 3x$
- (iv) $u(x, 4) = x^2$ for $0 \leq x \leq 4, 0 \leq y \leq 4$, correct to 2 decimal places.

(OR)

- b. Solve by Crank Nicholson method the equation $u_{xx} = 16u_t$ subject to $u(x, 0) = 0$; $u(0, t) = 0$ and $u(1, t) = 100t$ for one time step taking $h = 0.25$.

Reg. No.

B.Tech. DEGREE EXAMINATION, NOVEMBER 2016
Third Semester

15MA206 – NUMERICAL METHODS

(For the candidates admitted during the academic year 2015 – 2016 onwards)

Note:

- (i) Part - A should be answered in OMR sheet within first 45 minutes and OMR sheet should be handed over to hall invigilator at the end of 45th minute.
- (ii) Part - B and Part - C should be answered in answer booklet.

Time: Three Hours

Max. Marks: 100

PART – A (20 × 1 = 20 Marks)

Answer ALL Questions

1. Under the condition $f(a)$ and $f(b)$ having opposite signs and $a < b$, the first approximation of one of the root of $f(x) = 0$, by regula Falsi method is given by
(A) $\frac{af(b) + bf(a)}{f(b) + f(a)}$ (B) $\frac{af(b) - bf(a)}{f(b) - f(a)}$
(C) $\frac{af(a) - bf(b)}{f(a) - f(b)}$ (D) $\frac{af(a) + bf(b)}{f(a) + f(b)}$
2. Newton's formula converges if
(A) $|f'(x)f''(x)| < |f(x)|^2$ (B) $|f(x)f''(x)| < |f'(x)|^2$
(C) $|f(x)f'(x)| < |f''(x)|^2$ (D) $|f(x)f'(x)| > |f''(x)|^2$
3. Which one of the following methods is a direct method to solve a system of simultaneous linear equation?
(A) Newton's method (B) Gauss elimination method
(C) Gauss seidal method (D) Gauss Jacobi method
4. By the method of least squares the error committed in fitting a straight line is
(A) $E = \Sigma y - a \Sigma xy - b \Sigma y^2$ (B) $E = \Sigma y - a \Sigma xy^2 - b \Sigma y$
(C) $E = \Sigma y^2 - a \Sigma xy - b \Sigma y$ (D) $E = \Sigma y^2 - a \Sigma x^2 y - b \Sigma y$
5. n^{th} difference of a polynomial of degree n is
(A) Zero (B) Positive
(C) Negative (D) Constant
6. $\frac{\Delta}{\nabla} - \frac{\nabla}{\Delta}$ is equal to
(A) 0 (B) $\Delta - \nabla$
(C) $\nabla - \Delta$ (D) $\nabla + \Delta$
7. If $u_1 = 1, u_3 = 17, u_4 = 43$ and $u_5 = 89$, the value of u_2 is
(A) 5 (B) 10
(C) 12 (D) 15

8. When the values of the independent variables are not equally spaced then we apply?
 (A) Lagrange's interpolation formula (B) Newton's forward interpolation formula
 (C) Newton's backward interpolation formula (D) Central difference interpolation formula
9. Newton's forward difference formula to get the first derivative at $x = x_0$ is
 (A) $\frac{dy}{dx} = \frac{1}{h} \left[\Delta y_0 - \frac{1}{2} \Delta^2 y_0 + \frac{1}{3} \Delta^3 y_0 - \dots \right]$ (B) $\frac{dy}{dx} = \frac{1}{h} \left[\Delta y_0 + \frac{1}{2} \Delta^2 y_0 + \frac{1}{3} \Delta^3 y_0 + \dots \right]$
 (C) $\frac{dy}{dx} = \frac{1}{h} \left[\nabla y_0 - \frac{1}{2} \nabla^2 y_0 + \frac{1}{3} \nabla^3 y_0 - \dots \right]$ (D) $\frac{dy}{dx} = \frac{1}{h} \left[1 + \Delta y_0 + \frac{1}{2} \Delta^2 y_0 + \frac{1}{3} \Delta^3 y_0 + \dots \right]$
10. Simpson's three eighth rule can be applied only when the number of interval is
 (A) Odd (B) Even
 (C) Multiple of 3 (D) Prime
11. $\frac{dy}{dx}$ at $x = 3$ from the data given below

x	1	2	3
y	1	8	27

 is
 (A) 3 (B) 25
 (C) 0 (D) 1
12. The error in Trapezoidal rule is of order
 (A) $\frac{h^2}{12}$ (B) h^2
 (C) $\frac{h^3}{6}$ (D) $\frac{3h^3}{8}$
13. Which of the following formula is a particular case of Runge Kutta formula of second order?
 (A) Taylor's formula (B) Euler's modified formula
 (C) Milne's predictor formula (D) Adam's predictor formula
14. How many prior values are required to predict the next value in Milne's method?
 (A) 3 (B) 2
 (C) 4 (D) 1
15. Predictor corrector method are
 (A) Multistep method (B) Self-correcting method
 (C) Single-step method (D) Self-starting method
16. Using R-K method of fourth order, calculate Δy if $k_1 = 0.1$, $k_2 = .11$, $k_3 = .1105$ and $k_4 = .12105$
 (A) 0.1305 (B) 0.01515
 (C) 0.11034 (D) 0.1410
17. Classify $f_{xx} + 2f_{xy} + 4f_{yy} = 0$
 (A) Elliptic (B) Parabolic
 (C) Hyperbolic (D) Poisson

18. In solving $u_t = \alpha^2 u_{xx}$ by Crank Nicholson method, to simplify the explicit formula we take $\frac{(\Delta x)^2}{\alpha^2 k}$ as
 (A) $\frac{1}{2}$ (B) 2
 (C) 1 (D) 0
19. Bender-Schmidt recurrence equation is given by
 (A) $u_{i,j} = \frac{1}{4} [u_{i-1,j-1} + u_{i-1,j+1} + u_{i+1,j-1} + u_{i+1,j+1}]$ (B) $u_{i,j} = \frac{1}{4} [u_{i-1,j} + u_{i+1,j} + u_{i,j-1} + u_{i,j+1}]$
 (C) $u_{i,j+1} = u_{i-1,j} + u_{i+1,j} - u_{i,j+1}$ (D) $u_{i,j+1} = \frac{1}{2} [u_{i-1,j} + u_{i+1,j}]$
20. The solution of hyperbolic equation $u_{tt} = a^2 u_{xx}$ is stable if
 (A) $\lambda = \frac{1}{2}$ (B) $\lambda = \frac{1}{a}$
 (C) $\lambda = a$ (D) $\lambda = 1$

PART - B (5 × 4 = 20 Marks)

Answer ANY FIVE Questions

21. Fit a straight line to the data given below using least square method

x	0	1	2	3	4
y	1	1.8	3.3	4.5	6.3

22. Find an iterative formula for $\sqrt[3]{N}$, where N is a positive integer using Newton's method.
23. Represent the function $x^4 - 12x^3 + 42x^2 - 30x + 9$ in factorial notation with interval of differencing $h = 1$.
24. Calculate $\int_2^{10} \frac{dx}{1+x}$ by dividing the interval into eight equal parts.
25. Using Taylor's series method find y at $x = 0$, if $\frac{dy}{dx} = x^2 y - 1$, $y(0) = 1$.
26. Classify the partial differential equation $x^2 f_{xx} + (1 - y^2) f_{yy} = 0$.
27. Write down Laplace equation and its finite difference analogue and the standard five point formula.

PART - C (5 × 12 = 60 Marks)

Answer ALL Questions

28. a.i. Given the following data

x	0	1	2	3	4
y	1	5	10	22	38

Find the parabola of best fit using least square method.

- ii. Find the positive root of $x^3 - 2x - 5 = 0$ by Regula Falsi method.